

$$\begin{aligned}
[\rho(t_1)] &= \rho(t_1) && + 3\sigma_{\rho(t_1)} \cdot \delta\rho(t_1) \\
[\bar{A}z(t_1)] &= \bar{A}z(t_1) && + CI_{\bar{A}z(t_1)} \cdot \delta\bar{A}z(t_1) \\
[\bar{E}l(t_1)] &= \bar{E}l(t_1) && + CI_{\bar{E}l(t_1)} \cdot \delta\bar{E}l(t_1) \\
[\rho(t_3)] &= \rho(t_3) && + 3\sigma_{\rho(t_3)} \cdot \delta\rho(t_3) \\
[\bar{A}z(t_3)] &= \bar{A}z(t_3) && + CI_{\bar{A}z(t_3)} \cdot \delta\bar{A}z(t_3) \\
[\bar{E}l(t_3)] &= \bar{E}l(t_3) && + CI_{\bar{E}l(t_3)} \cdot \delta\bar{E}l(t_3)
\end{aligned} \tag{5}$$

where CI indicates the confidence interval for the measurements or search space for the optimization variables. We then perform steps 1–7 of Algorithm 1 with ADS. This operation produces

$$\Delta^T \Delta = \bigcup_{i=1}^{N_s} \mathcal{T}_{\Delta^T \Delta}^i(\mathbf{x}) \tag{6}$$

with $\mathbf{x} = (\rho(t_1), \bar{A}z(t_1), \bar{E}l(t_1), \rho(t_3), \bar{A}z(t_3), \bar{E}l(t_3))$ and N_s the number of domains generated by ADS. A minimization of the residual squares is then performed on each domain. Only those for which $\Delta^T \Delta$ is zero or below a certain thresholds are retained. These domains describe the set of spacecraft states compatible with the available observations, which can be propagated forward in time to predict future detections.

3.2 Accurate angular measurements on one tracklet

The second case considers the availability of accurate angular measurements on one of the two tracklets. For illustration, we consider these measurements together with the range and range-rate to be available on the first tracklet at t_1 . Range and range-rate are also available for the second tracklet, at t_2 . Concerning the angular data and their rates, we work with the regressed values and their rate. The procedure is summarized in Algorithm 2.

Algorithm 2: IOD with accurate angular measurements on a single tracklet

Input: Initial guesses for $\dot{\bar{A}z}(t_1), \dot{\bar{E}l}(t_1)$

Output: The values of $\dot{\bar{A}z}(t_1), \dot{\bar{E}l}(t_1)$ that solve $\Delta = \mathbf{0}$

1. Guess initial values for $\dot{\bar{A}z}(t_1), \dot{\bar{E}l}(t_1)$;
2. Take observations $\rho(t_1), \dot{\rho}(t_1), \bar{A}z(t_1), \bar{E}l(t_1)$;
3. Compute the Cartesian state:

$$\mathbf{r}(t_1)(\rho(t_1), \dot{\rho}(t_1), \bar{A}z(t_1), \bar{E}l(t_1), \dot{\bar{A}z}(t_1), \dot{\bar{E}l}(t_1))$$

$$\mathbf{v}(t_1)(\rho(t_1), \dot{\rho}(t_1), \bar{A}z(t_1), \bar{E}l(t_1), \dot{\bar{A}z}(t_1), \dot{\bar{E}l}(t_1));$$

4. Propagate to the state forward in time:

$$\mathbf{r}(t_2)(\rho(t_1), \dot{\rho}(t_1), \bar{A}z(t_1), \bar{E}l(t_1), \dot{\bar{A}z}(t_1), \dot{\bar{E}l}(t_1))$$

$$\mathbf{v}(t_2)(\rho(t_1), \dot{\rho}(t_1), \bar{A}z(t_1), \bar{E}l(t_1), \dot{\bar{A}z}(t_1), \dot{\bar{E}l}(t_1));$$

5. Build the residual vector Δ using the state at t_2 :

$$\Delta(\rho(t_1), \dot{\rho}(t_1), \bar{A}z(t_1), \bar{E}l(t_1), \dot{\bar{A}z}(t_1), \dot{\bar{E}l}(t_1)) = \begin{pmatrix} \rho^c(t_2) - \rho(t_2) \\ \dot{\rho}^c(t_2) - \dot{\rho}(t_2) \end{pmatrix};$$

6. Compute $\dot{\bar{A}z}(t_1), \dot{\bar{E}l}(t_1)$ such that $\Delta = \mathbf{0}$
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